

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

1220	11.647202200	fe80::6962:fd55:6f5...	fe80::6231:92ff:fe0...	DNS	96	Standard query 0xb59e HTTPS open.spotify.com
1221	11.647753400	fe80::6962:fd55:6f5...	fe80::6231:92ff:fe0...	DNS	96	Standard query 0x26f8 AAAA open.spotify.com
1222	11.652591100	fe80::6231:92ff:fe0...	fe80::6962:fd55:6f5...	DNS	194	Standard query response 0xb59e HTTPS open.spotify.com CNAME atc.spotify.map.fastly.net
1223	11.653429000	fe80::6231:92ff:fe0...	fe80::6962:fd55:6f5...	DNS	164	Standard query response 0x26f8 AAAA open.spotify.com CNAME atc.spotify.map.fastly.net

You can complete this as a practical Wireshark lab exercise. The exact packet numbers and IP addresses will be different on your computer.

Steps

1. Open **Wireshark**.
2. Select your active **Wi-Fi/Ethernet** interface.
3. Start capturing.
4. Open your browser and visit www.spotify.com.
5. Wait a few seconds for the page to load, then stop the capture.

1. Find DNS packets

Use this Wireshark filter:

- dns

Record any two DNS packets, for example:

Packet	Protocol	Information to Note
DNS Packet 1	DNS	Query for www.spotify.com
DNS Packet 2	DNS	Response containing the IP address for the requested domain

2. Find HTTPS packets

Spotify normally uses **HTTPS**, so use:

Packet	Protocol	Information to Note
HTTPS Packet 1	TLS	Client → Spotify server, encrypted HTTPS traffic
HTTPS Packet 2	TLS	Spotify server → client, encrypted response traffic

2. Simulate a DNS lookup for `www.flipkart.com` using the `nslookup` command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

```
C:\Windows\system32>nslookup flipkart.com
Server: UnKnown
Address: fe80::6231:92ff:fe04:38bd

Non-authoritative answer:
Name: flipkart.com
Address: 103.243.32.90
```

3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

Difference	HTTP	HTTPS
1. Security	Data is sent without encryption.	Data is encrypted using TLS.
2. Port	Usually uses port 80 .	Usually uses port 443 .
3. Data Protection	More vulnerable to eavesdropping and modification.	Helps protect data from being read or changed during transmission.

Real-world example

Instagram uses **HTTPS** to securely communicate between your browser/app and its servers. This helps protect information such as login credentials and other data while it travels across the Internet.

Simple example:

HTTP: Your device — readable data —> Website

HTTPS: Your device — encrypted data —> Website

4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

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5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a café).

Protocol: DHCP

Real-world scenario: Using Instagram at a café

When I connect my phone to a café's Wi-Fi and open Instagram, **DHCP (Dynamic Host Configuration Protocol)** automatically gives my phone the network information it needs.

How it works:

1. My phone connects to the café's Wi-Fi.
2. The DHCP server assigns the phone an **IP address**.
3. It also provides the **subnet mask, default gateway, and DNS server**.
4. My phone can now communicate through the Internet and access Instagram's servers.